



# CRYSTAL CALIBRATION SALES AND SERVICE CO., LTD.

45/48 Soi Salathammasop31, Salathammasop Rd.,

Salathammasop, Thawewatthana, Bangkok 10170 Thailand

Tel : 0-2408-8474-5 Fax : 0-2408-8477 Email : info@crystalcal.com www.crystalcal.com



## CERTIFICATE OF CALIBRATION

Certificate No. : 25-1390-027

Issue Date : 24 October 2025

Work Order No. : 25/1390

Customer Name : กลุ่มงานเทคนิคการแพทย์ โรงพยาบาลพระจอมเกล้า จังหวัดเพชรบุรี  
53 ถ.รถไฟ ต.คลองกระแซง  
อ.เมือง จ.เพชรบุรี 76000

Date of Received : 20 October 2025

Date of Calibration : 24 October 2025

Instrument Details : Description : Piston pipette (Auto pipette)  
Manufacturer : DLAB  
Model : N/A  
Serial No. : YM198AH0069492  
ID No. : ค18-ป001-01-1/63  
Volume range : 0.1  $\mu$ l to 2.5  $\mu$ l  
Type : Variable volume single channel (air displacement)  
Location : Mass and Volumetric Calibration Laboratory

Calibration Method : This instrument was calibrated by Gravimetric methods for the determination of measurement error according to calibration procedure no. CWI-V-01 based on ISO 8655 : 2002 Part 2, 6

### Environmental Condition

Ambient Temperature : Control at 20 °C  $\pm$  2.5 °C  
Relative humidity : Control at 50 %RH  $\pm$  10 %RH  
Atmospheric pressure : Monitoring at 1010 hPa  $\pm$  10 hPa

### Traceability of Measurement

: This certificate of calibration documents the traceability to national standard, which realize the unit of measurement according to the International system of Units (SI)

Calibrated by : Miss Nattharika Peedee  
Calibration Engineer

Approved by :   
( Mr. Anuwat Yaklermjit )  
Laboratory Manager

This certificate may not be reproduced other than in full except with the prior written approval of Crystal Calibration Sales and Service co., Ltd.

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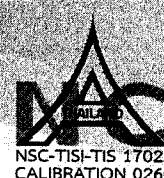


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**CERTIFICATE OF CALIBRATION**

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**Details of Calibration****1. Reference Standards Instrument**

| Instrument                          | Serial No. / ID No. | Certification   | Due Date         |
|-------------------------------------|---------------------|-----------------|------------------|
| 1.1 Electronic Balance              | BAL-06              | 25-0164-004     | 14 March 2026    |
| 1.2 Digital Thermometer with sensor | RTD-09              | 25-0164-009     | 3 March 2026     |
| 1.3 Thermo hygrometer               | TH-06               | 25-0164-012     | 3 March 2026     |
| 1.4 Barometer                       | BM-06               | L202509118-0001 | 9 September 2026 |

**2. Certificate traceable** : This certificate traceable to The International System of Unit refer to

1.1 Crystal Calibration Sales and Service Co., Ltd., NSC-ONSC Calibration No. 0260

1.2 to 1.3 Crystal Calibration Sales and Service Co., Ltd., NSC-ONSC Calibration No. 0260

1.4 Miracle international Technology Co., Ltd., NSC-ONSC Calibration No. 0052

**3. Condition of item** : Used**4. Calibration site** : Permanent**Result of Calibration**

This result of calibration : Without Adjustment

Resolution : 0.05  $\mu$ l

|                                |                   |         |         |         |
|--------------------------------|-------------------|---------|---------|---------|
| Test volume                    | ( $\mu$ l )       | 0.25    | 1.25    | 2.5     |
| Setting volume                 | ( $\mu$ l )       | 0.25    | 1.25    | 2.5     |
| Mean volume                    | ( $\mu$ l )       | 0.2172  | 1.2353  | 2.4587  |
| Uncertainty of measurement     | ( $\pm$ $\mu$ l ) | 0.013   | 0.014   | 0.014   |
| Coverage factor (k)            |                   | 2.05    | 2.05    | 2.05    |
| Systematic : Error ( $e_s$ )   | ( $\mu$ l )       | -0.0328 | -0.0147 | -0.0413 |
| : Error in percent ( % $e_s$ ) |                   | -1.314  | -0.589  | -1.651  |
| Random : Error ( $S_r$ )       | ( $\mu$ l )       | 0.012   | 0.013   | 0.012   |
| : Error in percent ( %CV )     |                   | 5.729   | 1.029   | 0.472   |

**Note :**

- Nominal volume piston pipette : 2.5  $\mu$ l
- The result of repeat the test 10 measurements
- The Calibration process calculate temperature reference is 20°C
- The values obtained are the mass values of the volume delivered (Ex.)
- Pipette tip : Manufacturer: eppendorf ; Color code: Colorless ; Capacity: 0.1  $\mu$ l to 10  $\mu$ l ; Length : 35 mm ; None filter
- The standard uncertainty of measurement has been determined in accordance with ISO/TR 20461:2000
- Calibrate items it good condition this report customer request and accepted in certificate

This result of calibration was found accurate as shown on date and place of calibration only.

The reported uncertainty of measurement was based on a standard uncertainty multiplied by a coverage factor  $k=2$  providing a level of confidence of approximately 95%.

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**Statements of conformity report**

Refer to Certificate No. : 25-1390-027

Issue Date : 24 October 2025

Work Order No. : 25/1390

**Detail of Equipment**

Description : Piston pipette (Auto pipette)

Manufacturer : DLAB

ID No. : ๑18-U001-01-1/63

Model : N/A

Volume range : 0.1 µl to 2.5 µl

Serial No. : YM198AH0069492

**Result of Calibration :**

This result of calibration : Without Adjustment

Resolution : 0.05 µl

|                                 |                                              |        |             |             |
|---------------------------------|----------------------------------------------|--------|-------------|-------------|
| Test volume                     | ( µl )                                       | 0.25   | 1.25        | 2.5         |
| Setting volume                  | ( µl )                                       | 0.25   | 1.25        | 2.5         |
| Mean volume                     | ( µl )                                       | 0.2172 | 1.2353      | 2.4587      |
| Uncertainty of measurement      | ( ± µl )                                     | 0.013  | 0.014       | 0.014       |
| Coverage factor (k)             |                                              | 2.05   | 2.05        | 2.05        |
| Systematic                      | : Error ( e <sub>s</sub> )                   | ( µl ) | -0.0328     | -0.0147     |
|                                 | : Error in percent ( % e <sub>s</sub> )      |        | -13.140     | -1.178      |
| Random                          | : Error ( S <sub>r</sub> )                   | ( µl ) | 0.012       | 0.013       |
|                                 | : Error in percent ( %CV )                   |        | 4.976       | 1.017       |
| Evaluate report<br>(Systematic) | : Abs. Error+Uncertainty                     | ( µl ) | 0.0461      | 0.0287      |
|                                 | : MPE. Error ( e <sub>s</sub> )              | ( µl ) | 0.0625      | 0.0625      |
|                                 | : Result (PASS / FAIL)                       |        | <u>PASS</u> | <u>PASS</u> |
|                                 | : Abs. Error+Uncertainty in percent          |        | 19.2644     | 2.3110      |
|                                 | : MPE. Error in percent ( % e <sub>s</sub> ) |        | 25          | 5           |
|                                 | : Result (PASS / FAIL)                       |        | <u>PASS</u> | <u>PASS</u> |
| Evaluate report<br>(Random)     | : MPE Error ( S <sub>r</sub> )               | ( µl ) | 0.05        | 0.05        |
|                                 | : Result (PASS / FAIL)                       |        | <u>PASS</u> | <u>PASS</u> |
|                                 | : MPE Error in percent ( CV )                |        | 20          | 4           |
|                                 | : Result (PASS / FAIL)                       |        | <u>PASS</u> | <u>PASS</u> |

**This statement judgment and decision by Laboratory guideline only**

The conformity certificate documents validity following ISO/IEC Guide 98-4 : Role of measurement uncertainty in conformity assessment based on statement with guard band refer to ISO 8655 : 2022 consider expanded measurement uncertainty (k=95%)

**The tolerance and decision rules ;**

MPE of Systematic error = Measurement uncertainty + Absolute error ; ≤ MPE is pass , &gt; MPE is Fail

MPE of Random error = Random error comparative with ISO 8655 : 2022 in table 1 ; ≤ MPE is pass , &gt; MPE is Fail

Statements of conformity decision by :

( Mr. Anuwat Yaklermjit )

Laboratory Manager